

Package ‘MEI’

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Type Package

Title Measurement Equivalence/Invariance Test

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This package provides a collection of functions for testing configural, metric, and scalar invariance, as well as comparing latent means. Applied to multiple-groups, multiple-occassions in longitudinal studies, multiple-sources in congruence research, and multilevel confirmatory factor analysis.

Notes:(1) Use 2,000 bootstrapped samples or one million sets of factor loadings and intercepts generated by Monte Carlo simulation based on the parameter estimates and the variance-covariance matrix of the freely estimated factor loadings and intercepts in the configural invariance model (or partial metric invariance model), and the confidence intervals of differences in factor loadings and intercepts between groups are calculated. (2) When more than two groups are tested for MEI, for each combination of reference item and argument, first compares each pair of groups using a defined adjusted Type I error rate, then applies the list-and-delete method to identify the set of invariant groups. (3) After identifying the sets of groups with invariant factor loadings or intercepts in each pair of reference and argument items, the model with the least number of freely estimated factor loadings and intercepts (the most parsimonious model with the largest number of invariant items) will be selected. (4) When multiple models have the same number of estimated parameters, the model with the best fit will be selected. (5) Missing values are handled with full-information maximum likelihood. (6) All non-converged and non-admissible boot-strapped samples are removed from the calculations. (7) All the identification of invariant sets is correct up to a maximum of 25 groups. If there are more than 12 sets of invariant groups, the remaining groups will be considered as non-invariant.

License GPL-3

Encoding UTF-8

LazyData TRUE

Depends R (>= 3.5.0),
lavaan (>= 0.6-15),
MASS (>= 7.3-60),

Imports MASS (>= 7.3-60),
lavaan (>= 0.6-15), parallel

URL <https://github.com/gche935/MEI>

BugReports <https://github.com/gche935/MEI/issues>

RoxygenNote 7.3.3

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CompareLoadings	<i>Metric Invariance Test</i>
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Description

Conduct a full metric invariance test and identify a partial metric invariance model for data from independent groups.

Usage

```
CompareLoadings(
  model,
  data.source,
  Groups,
  Cluster = "NULL",
  Bootstrap = 0,
  Type1 = 0.05,
  Type1Adj = "PFDR",
  BMSC = "SRMR"
)
```

Arguments

<code>model</code>	User-specified CFA model.
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Groups</code>	Grouping variable for cross-group comparisons.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>Bootstrap</code>	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
<code>Type1</code>	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.

Type1Adj	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all groups), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
BMSC	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).

Details

Reference: Measurement Equivalence/Invariance Test based on "Cheung, G. W. & Lau, R. S. (2012). A direct comparison approach for testing measurement invariance. *Organizational Research Methods*, 15, 167-198."

Value

Partial metric invariance model in the PMI.txt file.

Examples

```
## -- Example A: Measurement Invariance Test Across Groups -- ##

# Data file is "Example.A"

# Specify the measurement model - Model.A
Model.A <- '
    WorkLifeConflict =~ R45a + R45b + R45c + R45d + R45e
    Engagement =~ R90a + R90b + R90c
    Wellbeing =~ R87a + R87b + R87c + R87d + R87e
'

## Not run:
## ===== Full Measurement Invariance Test ===== ##
Full_MEI(Model.A, Example.A, Groups="Region")
## End (Not run)

## ===== Compare Loadings ===== ##
CompareLoadings(Model.A, Example.A, Groups="Region", Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")
```

CompareMeans

Scalar Invariance Test and Compare Latent Means

Description

Conduct scalar invariance test, identify a partial scalar invariance model, and compare latent means for data from independent groups.

Usage

```
CompareMeans(
  model.PMI,
  data.source,
  Groups,
  Cluster = "NULL",
  Bootstrap = 0,
  Type1 = 0.05,
  Type1Adj = "PFDR",
  BMSC = "SRMR",
  correct.cfi = TRUE
)
```

Arguments

<code>model.PMI</code>	Partial metric invariance model (PMI.Model.R from CompareLoadings() or user-specified).
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Groups</code>	Grouping variable for cross-group comparisons.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>Bootstrap</code>	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
<code>Type1</code>	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.
<code>Type1Adj</code>	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all groups), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
<code>BMSC</code>	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).
<code>correct.cfi</code>	If TRUE (default), baseline model will be corrected for calculation of incremental fit indices based on Widaman & Thompson (2003).

Details

Reference: Measurement Equivalence/Invariance Test based on "Cheung, G. W. & Lau, R. S. (2012). A direct comparison approach for testing measurement invariance. *Organizational Research Methods*, 15, 167-198."

Reference for `correct.cfi`: Widaman, K. F., & Thompson, J. S. (2003). On specifying the null model for incremental fit indices in structural equation modeling. *Psychological Methods*, 8(1), 16-37.

Value

Partial scalar invariance model in PSI.txt file and results of latent means comparisons.

Examples

```
# Data file is "Example.A"

# Specify the measurement model - Model.A
Model.A <- '
    WorkLifeConflict =~ R45a + R45b + R45c + R45d + R45e
    Engagement =~ R90a + R90b + R90c
    Wellbeing =~ R87a + R87b + R87c + R87d + R87e
'

## Not run:
## ===== Full Measurement Invariance Test ===== ##
Full_MEI(Model.A, Example.A, Groups="Region")

## ===== Compare Loadings ===== ##
CompareLoadings(Model.A, Example.A, Groups="Region", Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")
## End (Not run)

## ===== Compare Intercepts and Latent Means ===== ##
CompareMeans(PMI.Model.R, Example.A, Groups="Region", Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")
```

Full_MEI

Full Measurement Invariance Test

Description

Conduct configural invariance, full metric invariance, and full scalar invariance tests for data from independent groups.

Usage

```
Full_MEI(model, data.source, Groups, Cluster = "NULL", correct.cfi = TRUE)
```

Arguments

<code>model</code>	User-specified CFA model.
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Groups</code>	Grouping variable for cross-group comparisons.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>correct.cfi</code>	If TRUE (default), baseline model will be corrected for calculation of incremental fit indices based on Widaman & Thompson (2003).

Details

Reference for `correct.cfi`: Widaman, K. F., & Thompson, J. S. (2003). On specifying the null model for incremental fit indices in structural equation modeling. *Psychological Methods*, 8(1), 16-37.

Value

lavaan outputs, model fit for configural, metric and scalar invariance models, summary of fit statistics for each model.

Examples

```
## -- Example A: Measurement Invariance Test Across Groups -- ##

# Data file is "Example.A"

# Specify the measurement model - Model.A
Model.A <- '
    WorkLifeConflict =~ R45a + R45b + R45c + R45d + R45e
    Engagement =~ R90a + R90b + R90c
    Wellbeing =~ R87a + R87b + R87c + R87d + R87e
'

## ===== Full Measurement Invariance Test ===== ##
Full_MEI(Model.A, Example.A, Groups="Region")
```

 LGCompareLoadings

Metric Invariance for Longitudinal Panel Data

Description

Conduct metric invariance test and identify a partial metric invariance model for longitudinal panel data.

Usage

```
LGCompareLoadings(
  model,
  data.source,
  Cluster = "NULL",
  no.waves = 3,
  Bootstrap = 0,
  Type1 = 0.05,
  Type1Adj = "PFDR",
  BMSC = "SRMR"
)
```

Arguments

<code>model</code>	User-specified measurement model.
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>no.waves</code>	Number of waves for comparisons.
<code>Bootstrap</code>	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
<code>Type1</code>	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.

Type1Adj	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all waves), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
BMSC	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).

Details

Requires defining the measurement model only once. All indicators should have the suffix _T1, _T2 and _T3 (e.g., x1_T1, x1_T2, x1_T3) in the data file to indicate when the indicators were measured. Residuals of indicators are covaried across time automatically.

Value

Partial metric invariance model in the PMI.txt file.

Examples

```
## == Example B - Panel Data in Longitudinal Studies == ##

# Data file is "Example.B"

## Specify the measurement model - Model.B ##
Model.B <- '
  SWLS =~ x1 + x2 + x3 + x4 + x5
'

## ===== Compare Factor Loadings ===== ##
LGCompareLoadings(Model.B, Example.B, no.waves=3, Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")
```

LGCompareMeans	<i>Scalar Invariance Test and Compare Latent Means for Longitudinal Panel Data</i>
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Description

Conduct scalar invariance test, identify a partial scalar invariance model, and compare latent means for longitudinal panel data.

Usage

```
LGCompareMeans(
  model.PMI,
  data.source,
  Cluster = "NULL",
  no.waves = 3,
  Bootstrap = 0,
```

```

Type1 = 0.05,
Type1Adj = "PFDR",
BMSC = "SRMR",
correct.cfi = TRUE
)

```

Arguments

<code>model.PMI</code>	Partial metric invariance model (PMI.Model.R from LGCompareLoadings() or user-specified).
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>no.waves</code>	Number of waves for comparisons.
<code>Bootstrap</code>	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
<code>Type1</code>	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.
<code>Type1Adj</code>	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all waves), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
<code>BMSC</code>	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).
<code>correct.cfi</code>	If TRUE (default), baseline model will be corrected for calculation of incremental fit indices based on Widaman & Thompson (2003).

Details

Requires defining the measurement model only once. All indicators should have the suffix `_T1`, `_T2` and `_T3` (e.g., `x1_T1`, `x1_T2`, `x1_T3`) in the data file to indicate when the indicators were measured.

Residuals of indicators are covaried across time automatically.

Reference for `correct.cfi`: Widaman, K. F., & Thompson, J. S. (2003). On specifying the null model for incremental fit indices in structural equation modeling. *Psychological Methods*, 8(1), 16-37.

Value

Partial scalar invariance model in PSI.txt file and results of latent means comparisons.

Examples

```

## == Example B - Panel Data in Longitudinal Studies == ##

# Data file is "Example.B"

## Not run:
## Specify the measurement model - Model.B ##
Model.B <- '

```



```

    SWLS =~ x1 + x2 + x3 + x4 + x5
,

## ===== Compare Factor Loadings ===== ##
LGCompareLoadings(Model.B, Example.B, no.waves=3, Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")
## End (Not run)

## ===== Compare Means ===== ##
LGCompareMeans(PMI.Model.R, Example.B, no.waves=3, Type1=0.05, Type1Adj="PFDR", BMSC="SRMR")

```

MLCompareLoadings	<i>Metric Invariance Test Between Level 1 and Level 2</i>
-------------------	---

Description

Conduct configural invariance and full metric invariance test, and identify a partial metric invariance model for multilevel data.

Usage

```

MLCompareLoadings(
  model,
  data.source,
  Cluster = "NULL",
  Type1 = 0.05,
  Type1Adj = "PFDR",
  BMSC = "SRMR"
)

```

Arguments

model	User-specified measurement model.
data.source	A data frame containing the observed variables used in the model.
Cluster	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
Type1	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.
Type1Adj	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all groups), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
BMSC	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum sum of SRMR-within and SRMR-between values (default).

Details

Reference: Measurement Equivalence/Invariance Test based on "Cheung, G. W. & Lau, R. S. (2012). A direct comparison approach for testing measurement invariance. *Organizational Research Methods*, 15, 167-198."

Since all Level 1 (within-group) variables are group-mean centered, intercepts and latent means of all variables at Level 1 are set to zero. Hence, only configural and metric invariance across levels are tested. Bootstrapping is not available for a multi-level model.

Define the measurement model only once without specifying the level, and the function will compare the model across levels.

Value

Partial metric invariance model in PMI.txt file.

Examples

```
## == Example D - Multilevel Confirmatory Factor Analysis == ##
# Data file is "Example.D"; cluster variable is "ID"

## Specify the measurement model - Model.D ##
Model.D <- 'OLBI =~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8'

## ===== Compare Loadings ===== ##
MLCompareLoadings(Model.D, Example.D, Cluster="ID", Type1=0.05, Type1Adj=NULL, BMSC="SRMR")
```

PAIRCompareLoadings	<i>Metric Invariance Test Between Two Sources</i>
---------------------	---

Description

Conduct metric invariance test and identify a partial metric invariance model for data from two non-independent sources.

Usage

```
PAIRCompareLoadings(
  model,
  data.source,
  Cluster = "NULL",
  Bootstrap = 0,
  Type1 = 0.01,
  Type1Adj = "PFDR",
  BMSC = "SRMR"
)
```

Arguments

<code>model</code>	User-specified measurement model.
<code>data.source</code>	A data frame containing the observed variables used in the model.
<code>Cluster</code>	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
<code>Bootstrap</code>	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
<code>Type1</code>	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.
<code>Type1Adj</code>	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all waves), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
<code>BMSC</code>	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).

Details

Requires defining the measurement model only once. All indicators should have the suffix `_G1` and `_G2` (e.g., `x1_G1`, `x1_G2`, `x2_G1`, `x2_G2`) in the data file to indicate the source from which the indicators were measured.

Residuals of indicators are covaried across sources automatically.

Value

Partial metric invariance model in the PMI.txt file.

Examples

```
## == Example C - Non-independent Data from Two Sources == ##

# Data file is "Example.C"

## Specify the measurement model - Model.C ##
Model.C <- '
  External =~ x1 + x2 + x3 + x4 + x5 + x6
  Internal =~ x7 + x8 + x9 + x10
'

## ===== Compare Factor Loadings ===== ##
PAIRCompareLoadings(Model.C, Example.C, Type1=0.01, Type1Adj="PFDR", BMSC="SRMR")
```

PAIRCompareMeans	<i>Scalar Invariance Test and Compare Latent Means in Congruence/Fit/Agreement Studies</i>
------------------	--

Description

Conduct scalar invariance test, identify a partial scalar invariance model, and compare latent means for data from two non-independent sources.

Usage

```
PAIRCompareMeans(
  model.PMI,
  data.source,
  Cluster = "NULL",
  Bootstrap = 0,
  Type1 = 0.01,
  Type1Adj = "PFDR",
  BMSC = "SRMR",
  correct.cfi = TRUE
)
```

Arguments

model.PMI	Partial metric invariance model (PMI.Model.R from PAIRCompareLoadings() or user-specified).
data.source	A data frame containing the observed variables used in the model.
Cluster	Cluster variable for nested data. The Monte Carlo simulation method should be used for nested data.
Bootstrap	Number of bootstrap samples, must be between 500 and 10,000. If not specified, 1 million Monte Carlo simulated samples (default) will be used.
Type1	Overall Type I error rate (default is 0.05) for identifying non-invariant items in the List and Delete method.
Type1Adj	Adjustment of Type I error rate for multiple tests for each construct. Default is Possible False Discovery Rate (PFDR = Type I error rate/No. of freely estimated factor loadings/intercepts across all waves), can also use (BON)ferroni adjustment (Type I error rate/No. of pairwise comparisons), or "NULL".
BMSC	Best Model Selection Criterion used to select among models with the same number of estimated parameters. "ChiSquare" = minimum Chi-Square value, "CFI" = maximum CFI value, "RMSEA" = minimum RMSEA value, and "SRMR" = minimum SRMR value (default).
correct.cfi	If TRUE (default), baseline model will be corrected for calculation of incremental fit indices based on Widaman & Thompson (2003).

Details

Requires defining the measurement model only once. All indicators should have the suffix _G1 and _G2 (e.g., x1_G1, x1_G2, x2_G1, and x2_G2) in the data file to indicate from which source the indicators were measured.

Residuals of indicators are covaried across sources automatically.

Reference for correct.cfi: Widaman, K. F., & Thompson, J. S. (2003). On specifying the null model for incremental fit indices in structural equation modeling. *Psychological Methods*, 8(1), 16-37.

Value

Partial scalar invariance model in PSI.txt file and results of latent means comparisons.

Examples

```
## == Example C - Non-independent Data from Two Sources == ##

# Data file is "Example.C"

## Not run:
## Specify the measurement model - Model.C ##
Model.C <- '
    External =~ x1 + x2 + x3 + x4 + x5 + x6
    Internal =~ x7 + x8 + x9 + x10
'

## ===== Compare Factor Loadings ===== ##
PAIRCompareLoadings(Model.C, Example.C, Type1=0.01, Type1Adj="PFDR", BMSC="SRMR")
## End (Not run)

## ===== Compare Means ===== ##
PAIRCompareMeans(PMI.Model.R, Example.C, Type1=0.01, Type1Adj="PFDR", BMSC="SRMR")
```

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